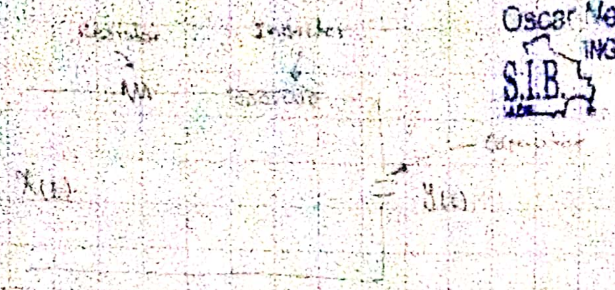
materia: fecha:

Modelo Matemático



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Voltaje entrada $X(t)$

$$X(t) = V_R + V_L + V_C$$

$$X(t) = R\,i(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\,dt$$

$$\mathcal{L}\{X(t)\} = R\,\mathcal{L}\{i(t)\} + L\,\mathcal{L}\left\{\frac{di(t)}{dt}\right\} + \frac{1}{C}\,\mathcal{L}\left\{\int i(t)\,dt\right\}$$

$$X(s) = R\,I(s) + Ls\,\mathcal{L}\{i(t)\} + \frac{1}{C}\,\frac{I(s)}{s}$$

$$X(s) = R\,I(s) + Ls\,I(s) + \frac{I(s)}{Cs}$$

$$y(s) = \frac{I(s)}{Cs}$$

Finalmente:

$$FT = \frac{\frac{I(s)}{Cs}}{R\,I(s) + Ls\,I(s) + \frac{I(s)}{Cs}} \rightarrow FT = \frac{I(s)\left(\frac{1}{Cs}\right)}{I(s)\left(R + Ls + \frac{1}{Cs}\right)}$$

$$\rightarrow FT = \frac{\frac{1}{Cs}}{R + Ls + \frac{1}{Cs}} \rightarrow FT = \frac{1}{Cs\left(R + Ls + \frac{1}{Cs}\right)}$$

$$\rightarrow FT = \frac{1}{CsR + Ls^2 + 1}$$

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materia: fecha:

$$V_{i(t)} = 2V_R + 2V_L + V_C$$

$$2R\,i_{(t)} + 2L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\,dt$$

$$= 2R\,\mathcal{L}\{i_{(t)}\} + 2L\,\mathcal{L}\left\{\frac{di(t)}{dt}\right\} + \frac{1}{C}\mathcal{L}\left\{\int i(t)\,dt\right\}$$

$$= 2R I_{(s)} + 2L_s\{I(s)\} + \frac{1}{C}\frac{I(s)}{s}$$

$$V_{i(s)} = 2R I_{(s)} + 2L_s I_{(s)} + \frac{I(s)}{Cs}$$

$$V_{o(s)} = L_s I_s$$

$$FT = \frac{L_s I_s}{2R I_{(s)} + 2L_s I_{(s)} + \frac{I_{(s)}}{Cs}} \rightarrow FT = \frac{L_s I_s}{I_{(s)}\left(2R + 2L_s + \frac{1}{Cs}\right)}$$

$$FT = \frac{L_s}{2R + 2L_s + \frac{1}{Cs}} \rightarrow FT = \frac{L_s}{2RC_s + 2L_sC_s + 1}$$

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